ICT305 Topics in IT

Topic 1
Artificial Intelligence and Machine Learning

CIHE Australia

Unit Introduction

The unit discusses topics surrounding innovation and emerging trends that are becoming important in the field of Information Technology. Topics include:

- AI and Machine Learning,
- Internet of Things (IoT),
- Data Mining,
- Web 3
- Virtual Reality (VR) and Augmented Reality (AR),
- Internet Information Systems and Mobile Computing Platforms.
- Quantum Computing
- Robotic Process Automation (RPA), and Smart Cities

Students will explore each of these the topics and get a broad overview to be able to get functioning knowledge in each of the area. This will allow students to be able to specialize in future in any one of the chosen topics.

Artificial Intelligence (AI): Initial Appearances

- **The first generation of AI researchers made these predictions about their work**:
  - *1958, H. A. Simon and Allen Newell*: "within ten years a digital computer will be the world's chess champion" and "within ten years a digital computer will discover and prove an important new mathematical theorem.“
  - *1965, H. A. Simon*: "machines will be capable, within twenty years, of doing any work a man can do.“
  - *1967, Marvin Minsky*: "Within a generation ... the problem of creating 'artificial intelligence' will substantially be solved.“
  - *1970, Marvin Minsky (in Life Magazine)*: "In from three to eight years we will have a machine with the general intelligence of an average human being."

AI: How to Define AI?

- The term got coined by John McCarthy in 1956 when a group of when scientists began exploring how computers could solve problems on their own.
  - **Def. 1** by David Marr: "*AI is the study of complex information processing problems that often have their roots in some aspects of biological information processing. The goal of the subject is to identify solvable and interesting information processing problems and solve them.*"
  - **Def. 2** by Rodney Brooks: "*The intelligent connection of perception to action.*"
  - **Def. 3** by Alan Turing: "*Actions that are indistinguishable from a human's ones.*"

What is Knowledge and Intelligence?

- **Knowledge** is a theoretical or practical understanding of a subject or a domain.
- **Knowledge** is the sum of what is currently Known.
- Someone's **intelligence** is their ability to understand and learn things.
- **Intelligence** is the ability to think and understand instead of doing things by instinct or automatically.
- It gives some flexibility.
  - It does not specify whether it is some**one** or some**thing** that has the ability to ***think*** and ***understand***.

What is Thinking and Intelligence?

- **Thinking** is the activity of using your brain to consider a problem or to create an idea.
- In order to think, some**one** or some**thing** has to have a brain – an organ that enables some**one** or some**thing** to learn and understand things, to solve problems and to make decisions.
- So, we can now define Intelligence:

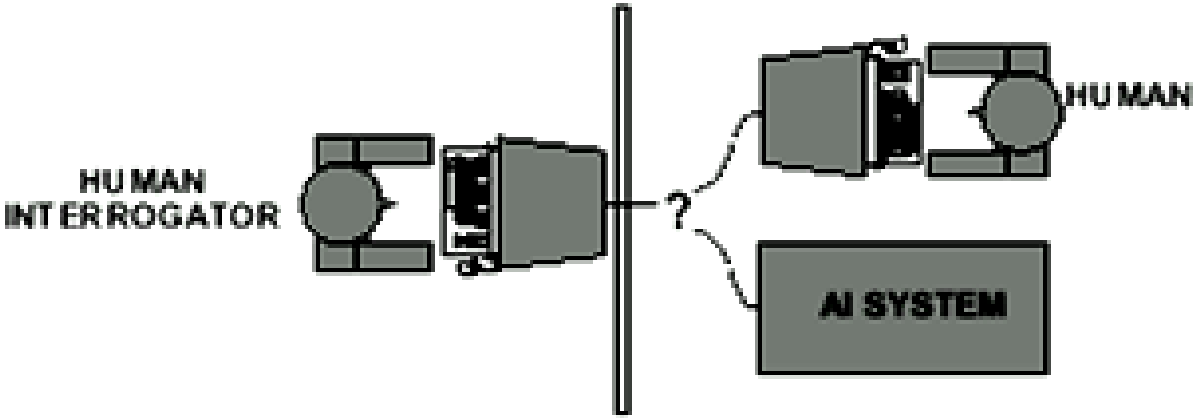
"It is the ability to learn and understand, to solve problems and to make decisions."

Turing Machine and Turing Test

- **Turing Test**:
  - The Turing Test evaluates whether a machine can exhibit intelligent behavior equivalent to a human.
  - In the test, a human interrogator engages in text-based conversations with two hidden participants - one human and one computer.
  - If the interrogator cannot reliably identify which participant is the computer, the machine is said to have passed the test.
- **How does the machine with thinking ability perform and how to test it?**

Acting Humanly: The Full Turing Test

- Alan Turing's 1950 article *Computing Machinery and Intelligence* discussed conditions for considering a machine to be intelligent
  - "Can machines think?" ⟷ "Can machines behave intelligently?"
  - The Turing test (The Imitation Game): Operational definition of intelligence.



Turing Imitation game

The Turing Test

http://aimovie.warnerbros.com

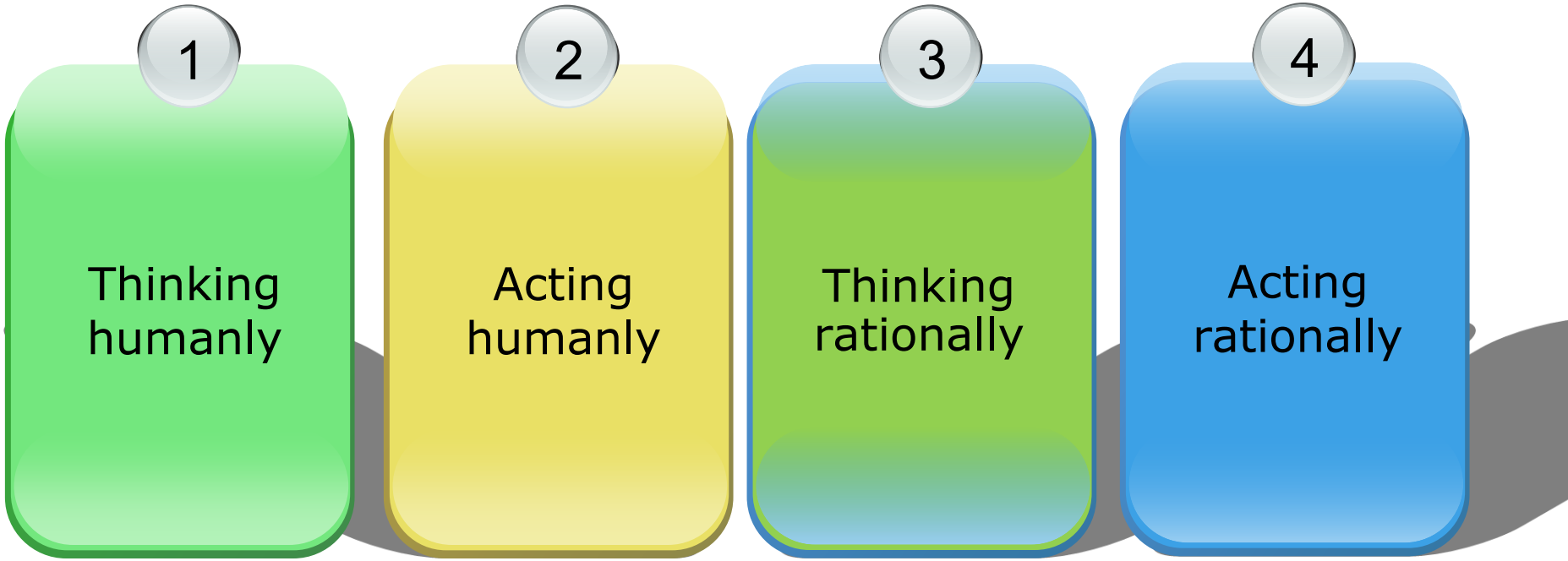
http://www.ai.mit.edu/projects/infolab/

Intelligence Machine (Cont.)

- A machine is thought intelligent if it can achieve human-level performance in some situations.
- To build an intelligent machine, we have to capture, organize and use human expert knowledge in some problem area.

What is AI?

Views of AI fall into four categories:



AI: Properties and Characteristics of AI

- **Automation**: Executes tasks without human intervention.
- **Learning & Adaptation**: Improves performance through data and experience.
- **Reasoning & Problem-Solving**: Analyzes data and makes decisions logically.
- **Perception**: Interprets sensory data (e.g., images, sounds).
- **Natural Language Processing** (NLP): Understands and generates human language.
- **Pattern Recognition**: Identifies patterns in large datasets.
- **Generalization**: Applies learned knowledge to new situations.
- **Autonomy**: Operates independently with minimal human input.
- **Robustness**: Functions effectively despite challenges.
- **Ethical Decision-Making**: Considers fairness, transparency, and ethics.
- **Creativity**: Generates new ideas, designs, and content.
- **Scalability**: Handles large-scale data and computations.
- **Collaboration**: Works alongside humans to enhance decision-making.

AI: Properties and Characteristics of AI

- **AI Levels**:
  - ***Narrow AI***: An artificial intelligence is said to be narrow when the machine can perform a specific task better than a human. The current research of AI is here now.
  - ***General AI***: An artificial intelligence reaches the general state when it can perform any intellectual task with the same accuracy level as a human would.
  - ***Strong AI***: An AI is strong when it can beat humans in many tasks.
- **Major Goals**:
  - **Understand** the principles that make intelligence possible (in humans, animals, and artificial agents).
  - **Develop** intelligent machines or agents (no matter whether they operate as humans or not).
  - **Formalize** knowledge and mechanizing reasoning in all areas of human endeavor.
  - **Make** the working with computers as easy as working with people.
  - **Develop** human-machine systems that exploit the complementariness of human and automated reasoning.

AI: Different Views on AI

- **Philosophy, Ethics, and Religion**:
  - What defines intelligence?
  - Can we formally express it?
  - How do we define the mind as a machine?
- **Cognitive Science, Neuroscience, Psychology, and Linguistics**:
  - Understand natural intelligence.
  - Learn principles of intelligent behavior.
- **Engineering**:
  - Can we build intelligent systems?
  - Focus on autonomous systems that replicate or enhance human capabilities.
- **How should scientists from different areas of science view AI and what technical elements (i.e., models, libraries, and etc.) should we have inside AI?**

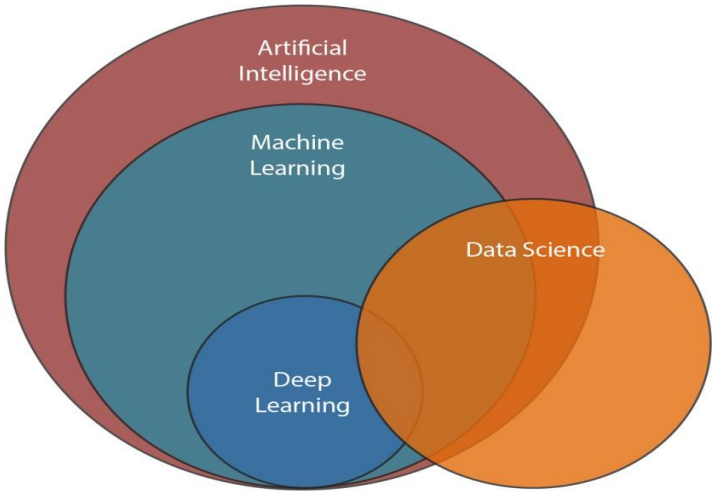
AI: What is Inside AI?

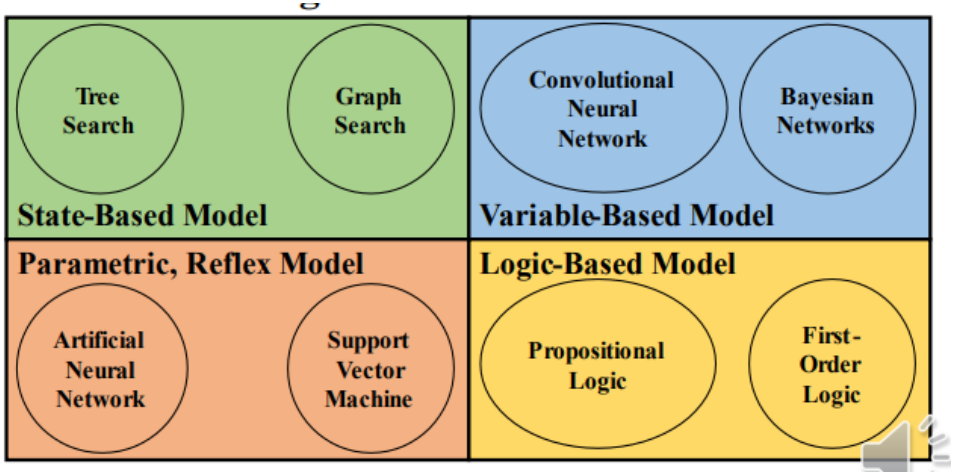
- **Applications**:
  - ✓ Image and Speech Recognition
  - ✓ Natural Language Processing
  - ✓ Autonomous Driving

- **Software/Hardware**:
  - ✓ Graphical Processing Unit
  - ✓ Parallel Processing Tools (e.g. Spark)
  - ✓ Cloud Data Storage and Computing System

- **Types of Models**:
  - ✓ Rule-based Models
  - ✓ Machine Learning
  - ✓ Deep Learning

- **Programming Languages and Libraries**:
  - ✓ Python, MATLAB, Java, and C++
  - ✓ TensorFlow, Keras, PyTorch, OpenCV, and Caffe





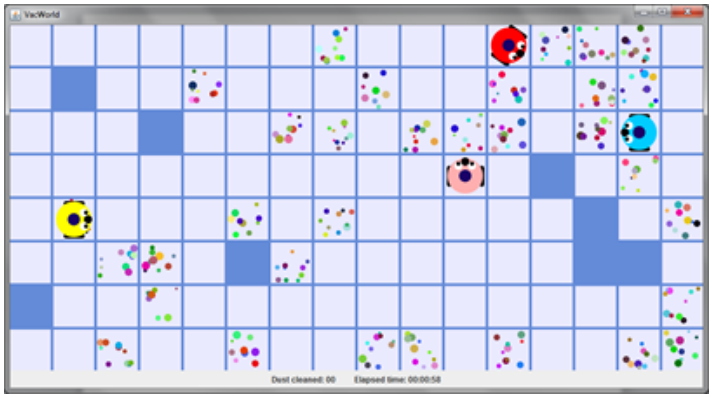
AI: Models to Study - 1

- *The array of problems the businesses face is huge, and the variety of models used to solve these problems is quite wide, as some algorithms are better at dealing with certain types of problems than the others.*
- *One needs a clear understanding of what every type of models is good for.*
- **State-Based Models**:
  - ***Solutions*** are defined as a sequence of steps.
  - ***Model*** a task as a graph of states and a solution as a path in the graph.
  - ***A state*** captures all of the relevant information about the past in order to act in the future.
  - **Apps**: Navigation and Games.
  - **Options**:
    - Tree Search (Breadth-first search, Depth-first search, and Iterative deepening),
    - Graph search (Dynamic programming)

Example Problem: Vacuum World

This can be formulated as a problem as follows:

- **States**:
  - The agent is in one of two locations, each of which might or might not contain dirt. For n location, 2^n states.
- **Initial State**:
  - Any state can be designated as the initial state
- **Successor Function**:
  - This generates the legal states that result from trying the three actions (Left, Right, Suck).
- **Goal Test**:
  - This cheeks whether all square is clean.
- **Path Cost**:
  - Each step costs 1, so the path cost is the number of steps in the path.



Types of Search Strategies

- **Uninformed Search Strategies:**
  - The algorithm are **uninformed**, in the sense that they are given no information about the problem other than its definition.
  - These strategies can find solutions to problem by systematically generating new states and testing them against the goal.
  - Uninformed search strategies use only the information available in the problem definition.
  - They can only generate successors and distinguish a goal state from a nongoal state.
- **Informed Search Strategies:**
  - **Informed search algorithm** have some idea of where to look for solutions.
  - This uses problem specific knowledge and can find solutions more efficiently.

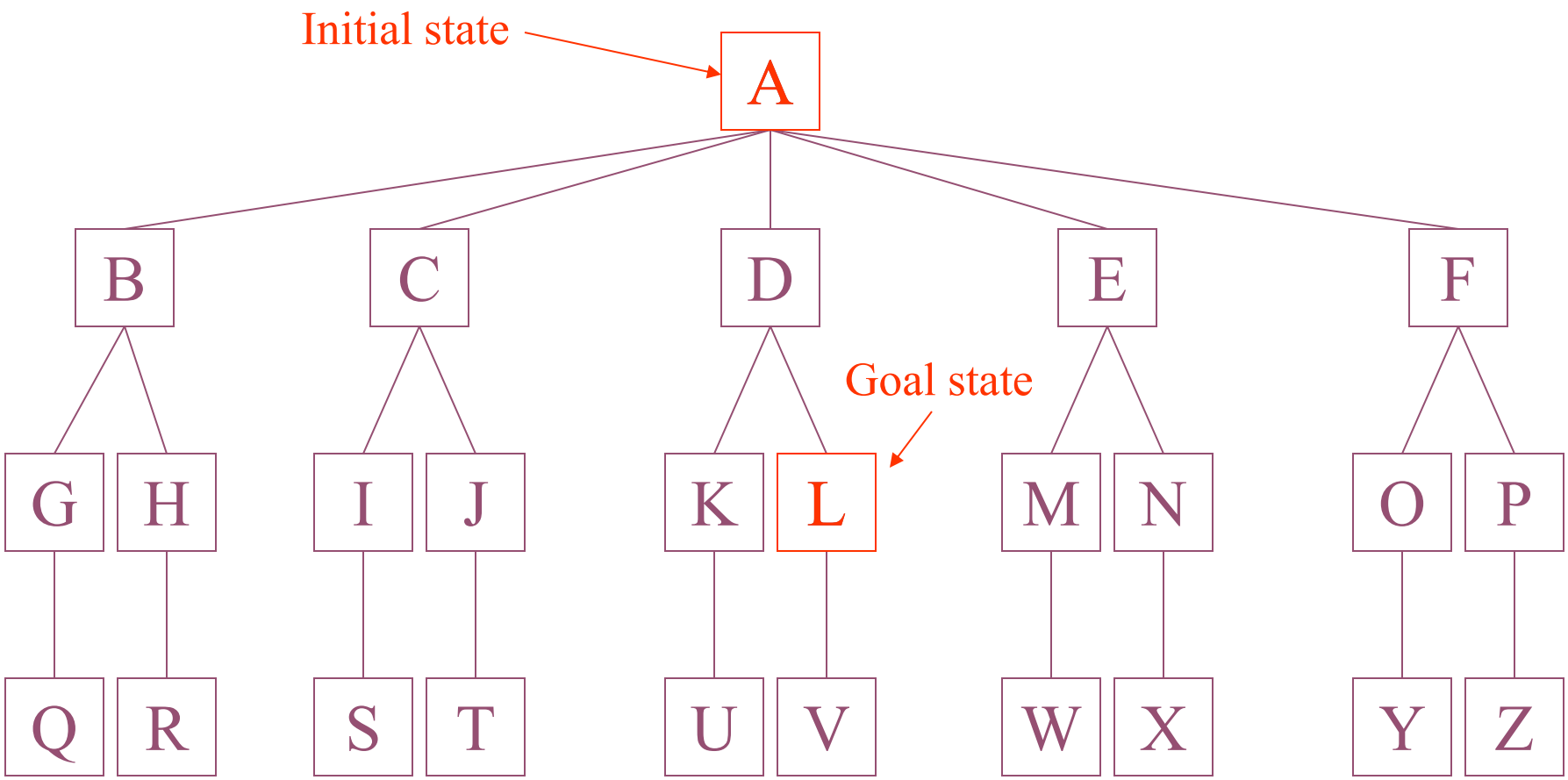
Uninformed Search Strategies

- Breadth – first Search
- Uniform – cost Search
- Depth - first Search
- Depth - limited Search
- Iterative deepening Depth-first Search

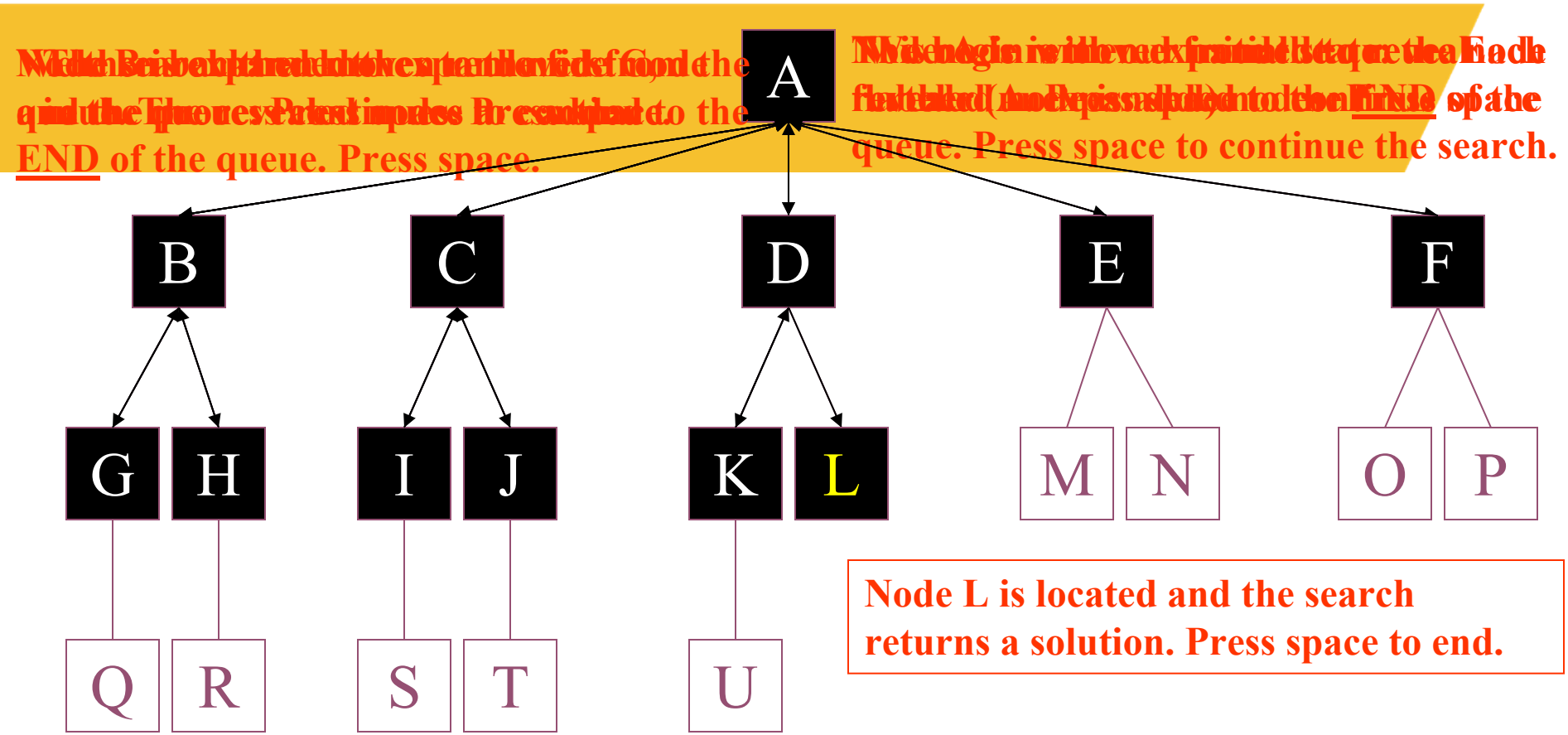
BFS: Definition

- **BFS** is a simple strategy in which root node is expanded first.
- Then all the successors of the root node are expanded and the same process repeats.
- Uses 'Queue' to ensure 'first-in-first-out' requirements.

The example node set



Press space to see a BFS of the example node set



Press space to begin the search

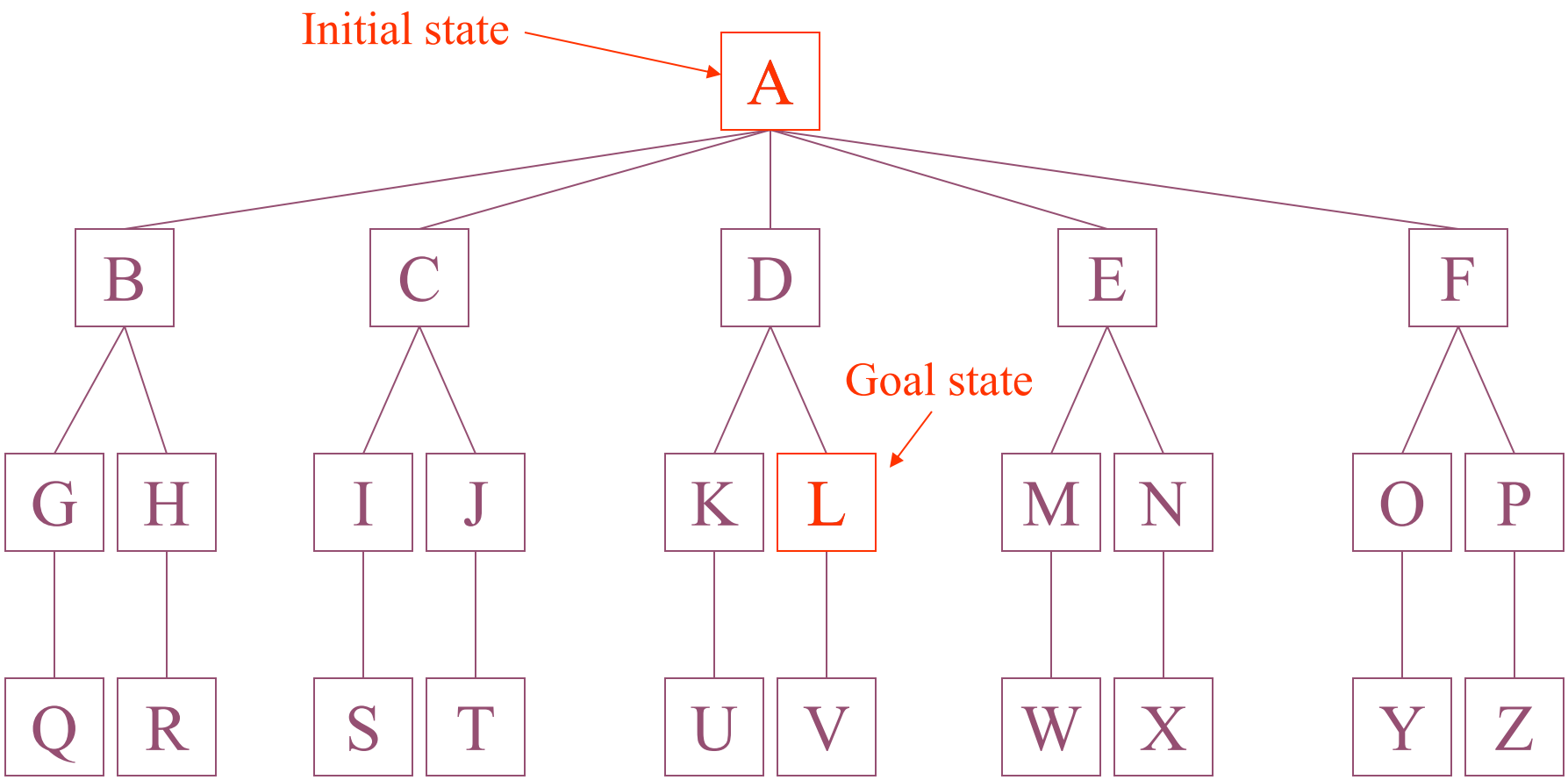
Size of Queue: 0	**Queue: Empty**	
Nodes expanded: 11	**FINISHED SEARCH**	**Current level: 2**

BREADTH-FIRST SEARCH PATTERN

Depth-First Search

- **Depth first search** explores one branch of a tree before it starts to explore another branch.
- It can be implemented by adding newly expanded nodes at the **front** of the queue.
- Using the general search algorithm we can implement it as follows:
  - **Function** DEPTH-FIRST-SEARCH(problem) **returns** a solution or failure
  - **Return** GENERAL-SEARCH(problem, ENQUEUE-AT-FRONT)
- The following slide illustrates a depth-first search of 26 nodes (states) with an initial state of node A and a goal state of node L. Press space to see the example node set.

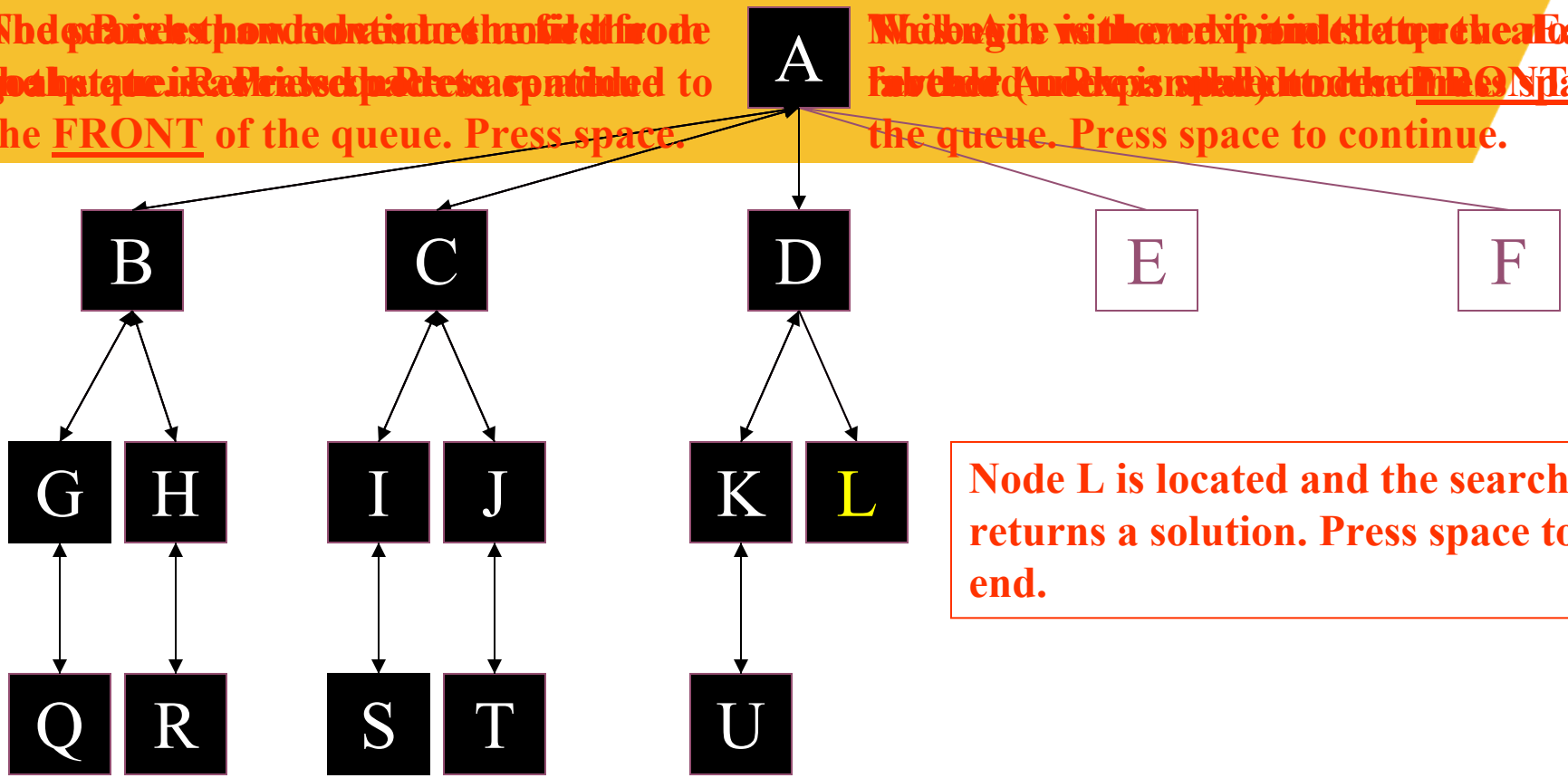
The example node set



Press space to see a DFS of the example node set

The operation removes the first node from the queue. Press space to continue.

Each node is added to the FRONT of the queue. Press space to continue.



Node L is located and the search returns a solution. Press space to end.

Press space to begin the search

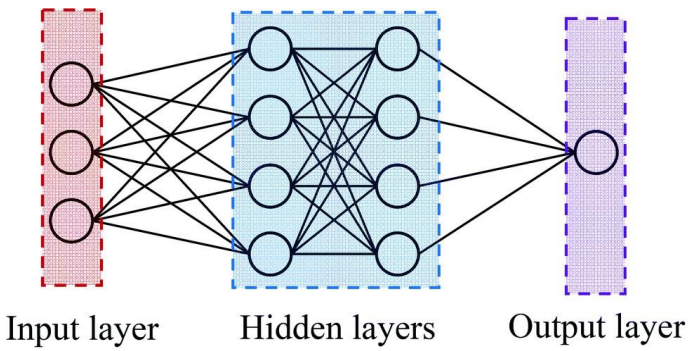
Size of Queue: 0	Queue: Empty	
Nodes expanded: 14	FINISHED SEARCH	Current level: 2

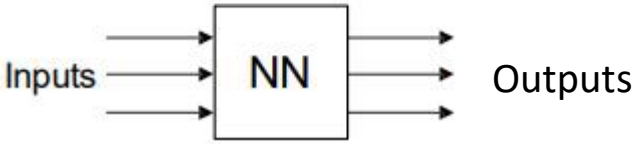
DEPTH-FIRST SEARCH PATTERN

AI: Models to Study - 2

Parametric, Reflex-Based Models:

- Given a set of <Input, Output> pairs of training data, learn a set of parameters that will map input to output for future data.
- **Apps**: Classification and Regression.
- **Key Techniques**:
  - Artificial Neural Networks (ANN),
  - Decision Trees,
  - Support Vector Machines,
  - Regression,
  - Principal Component Analysis,
  - K-Means Clustering,
  - K-Nearest Neighbor.





AI: Models to Study - 2

Variable-Based Models (Uncertainty):

- ***Concept***: The solution involves finding the right values for different variables in a problem.
- ***Examples***:
  - Sudoku: Filling in the numbers correctly.
  - Speech Recognition: Understanding spoken words.
  - Face Recognition: Identifying people in photos.

1	2	3
6		4
7	8	9

- **Techniques**:
  - **Convolutional Neural Networks (CNNs)**: Great for recognizing images and patterns.
  - **Constraint Satisfaction**: Solving puzzles by meeting specific rules.
  - **Bayesian Networks**: Making decisions with uncertain information.
  - **Factor Graphs**: Breaking problems into smaller parts.
  - **Dynamic Ordering**: Choosing the best order to solve problems.
  - **Hidden Markov Models (HMMs):** Predicting sequences, like words in speech.

AI: Models to Study - 3

Logic Based Model

- ***Concept***: Using **symbols** to represent different types of objects and applying logical rules to draw conclusions.
- ***Examples***:
  - Question Answering Systems: Finding answers to questions.
  - Natural Language Understanding: Making sense of human language.
- **Techniques**:
  - Propositional Logic: Basic true/false statements.
  - First-Order Logic: More detailed statements with objects and their relationships.
  - Knowledge Base: Storing facts and rules to use in reasoning.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

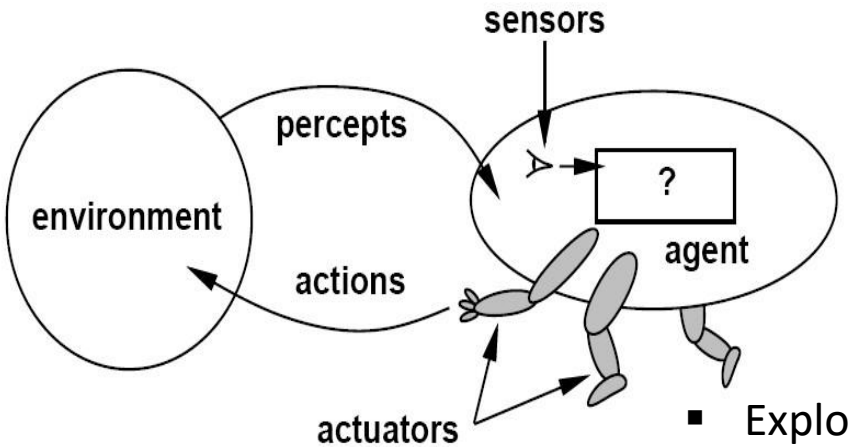
Logic Based Model

- **A and B (A ∧ B):**
  - This represents a logical conjunction.
  - The expression "A and B" is true only if both A and B are true.
  - Otherwise, "A and B" is false
  - For example: A = "Temperature is below 18°C", B = "House is occupied" -> **Action**: Turn on the heating.
- **A or B (A ∨ B):**
  - This represents a logical disjunction.
  - The expression "A or B" is true if at least one of A or B is true.
  - If both A and B are false, then "A or B" is false.
  - For example: A = "It is dark outside", B = "Someone is in the room" -> **Action**: Turn on the lights.
- **A -> B (A implies B or A → B):**
  - This represents a logical implication.
  - The expression "A -> B" is false only if A is true and B is false.
  - In all other cases (A is false, or B is true), "A -> B" is true.
  - For example: A = "Window is open", B = "It is cold outside" -> **Action**: Send a notification to close the window.

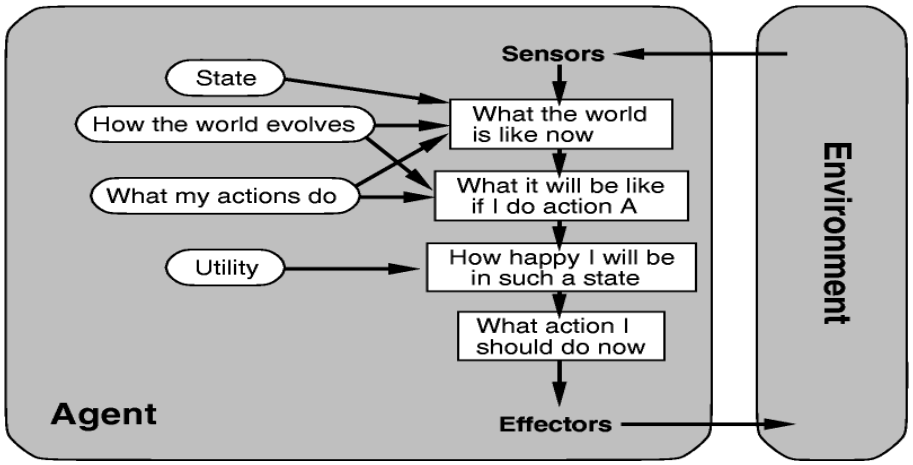
AI: What is an intelligent agent?

- An **Intelligent Agent** is a system:
  - It is an **AI-based Computing Engine** that includes an **AI Model** and is the core of **Intelligent Systems**.
  - ***Perception***: It observes and understands its environment. This environment could be the physical world, a user interacting through a screen, other agents, the internet, or a complex system.
  - ***Reasoning***: It thinks about what it perceives to interpret the situation, make decisions, solve problems, and figure out what to do next.
  - ***Action***: It takes actions in the environment to achieve the goals or tasks it was created for.
- **Characteristic Features:**
  - Knowledge Representation and Reasoning.
  - Transparency and Explanations.
  - Ability to Communicate.
  - Use of huge amounts of knowledge.

AI: What is an intelligent agent?



- Exploration of huge search spaces.
- Use of heuristics.
- Reasoning with incomplete or conflicting data.
- Ability to learn and adapt.

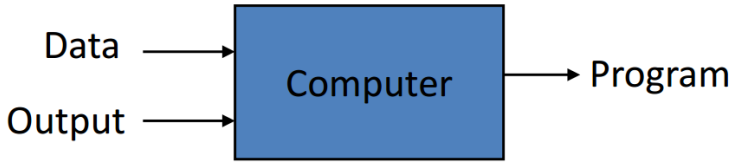


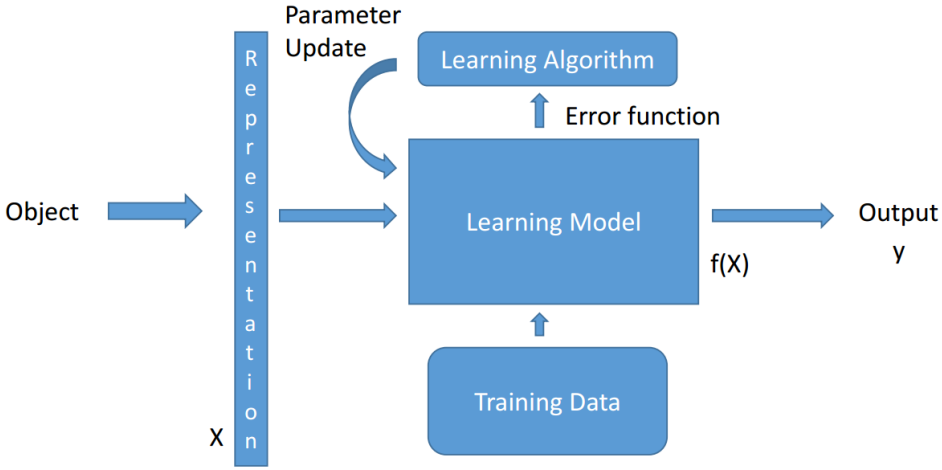
AI: Machine Learning – Definition

- **What is Machine Learning?**
  - A field of AI focused on creating computer systems that can get better at tasks by learning from data or their own experiences.
- **What is Learning?**
  - *"Learning is any process by which a system improves performance from experience."*

Traditional Programming

Machine Learning





Data to Model

Learning Algorithms

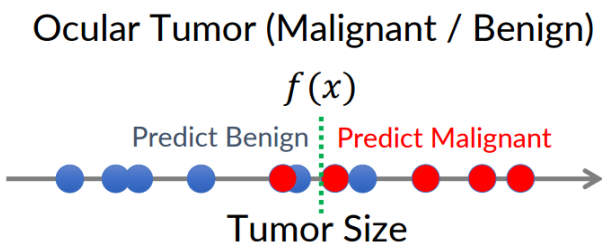
- **Supervised Learning**
  - **Definition**: In supervised learning, the model is trained on a labeled dataset, which means that each training example is paired with an output label.
    - The model learns to make predictions or decisions based on this labeled data.
  - **Example**: Classifying emails as "spam" or "not spam" using a dataset where emails are already labeled as such.
- **Unsupervised Learning**
  - **Definition**: In unsupervised learning, the model is given data without any explicit labels.
    - The goal is to find hidden patterns or intrinsic structures within the data, such as grouping similar items together.
  - **Example**: Grouping customers into different segments based on purchasing behavior, without predefined categories.

Learning Algorithms

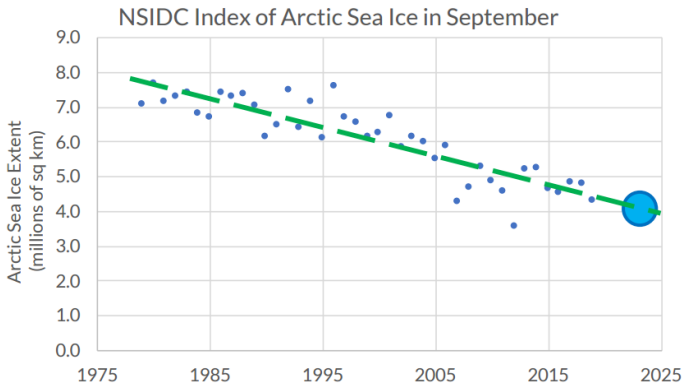
- **Semi-supervised**
  - Supervisory output available for few data points
  - Output not available for most data points
- **Reinforcement learning**
  - *Input*: Sequence of interactions with an environment
  - *Output*: Policy that performs a desired task

Machine Learning Models

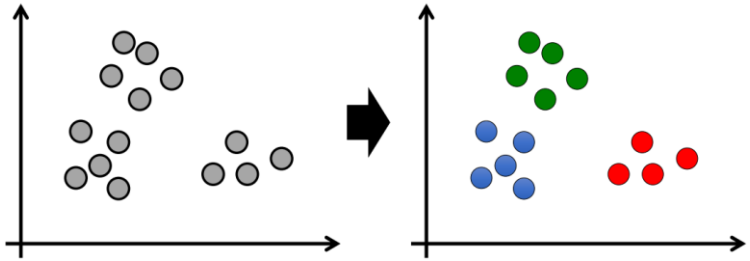
- **Classification**
  - Predicts category of input objects – predefined classes
  - Object recognition in images, email spam detection



- **Regression**
  - Predicts real valued output for a given input
  - Predicting value of a stock, predicting number of clicks in an advertisement



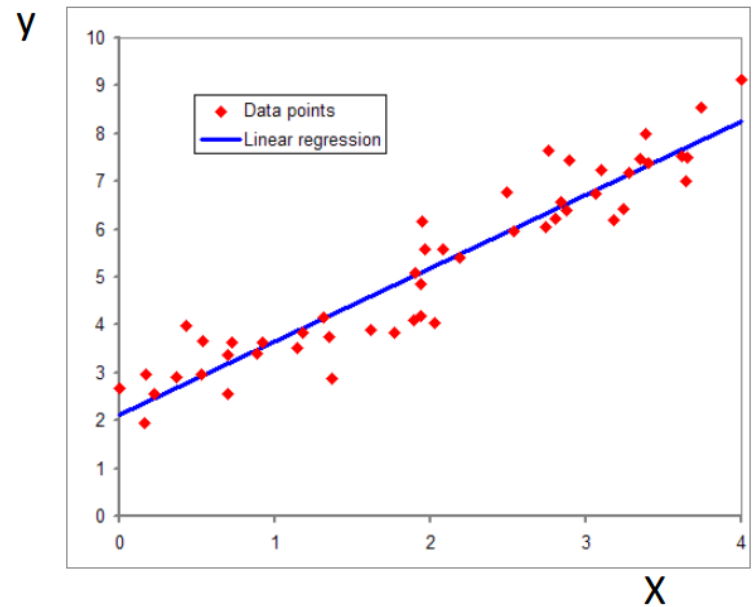
- **Clustering**
  - Groups objects into homogeneous clusters – clusters not predefined
  - Market segmentation, anomaly detection in industrial plants



Examples of Machine Learning Models

- **Classification and Regression**
  - Linear and Logistic Regression
  - Bayesian learning
  - K-Nearest neighbour
  - Decision Tree
  - Support Vector Machine
  - Boosting – Random Forests, Xgboost
  - Neural Networks and Deep Learning
- **Clustering**
  - K-means clustering
  - Hierarchical clustering
  - DBSCAN

Linear Regression



- Linear function in 1D: $y = b \times x$

- The space of linear functions $f(\beta, x)$

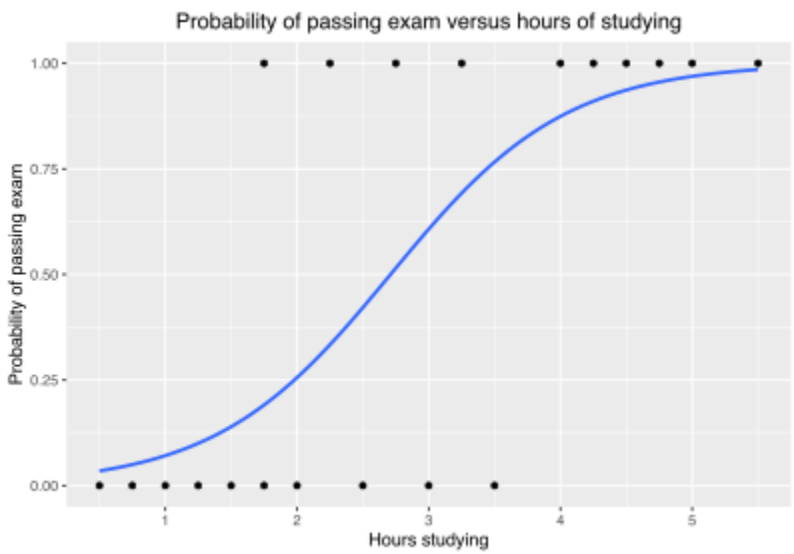
$$f_\beta(x) = \beta^\top x = [\beta_1 \quad \cdots \quad \beta_d] \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = \beta_1 x_1 + \cdots + \beta_d x_d$$

  - $x \in \mathbb{R}^d$ is called an **input** (a.k.a. **features** or **covariates**)
  - $\beta \in \mathbb{R}^d$ is called the **parameters** (a.k.a. **parameter vector**)
  - $y = f_\beta(x)$ is called the **label** (a.k.a. **output** or **response**)

Logistic Regression: Binary Classification

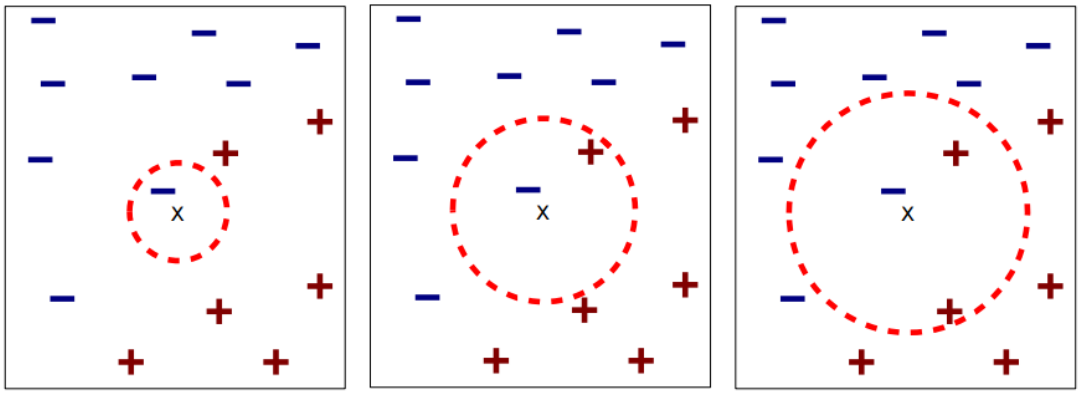
- **Logistic regression** is a statistical method used for binary classification problems, where the outcome is a categorical variable with two possible outcomes
- **Instead of modelling y**, model $P(y = 1|X) = p_x = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\cdots+\beta_n X_n)}}$
- **Predict** if a student will pass an exam depending on how many hours she has studied.

Hours (x_k)	0.50	0.75	1.00	1.25	1.50	1.75	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	4.00	4.25	4.50	4.75	5.00	5.50
Pass (y_k)	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1



K Nearest Neighbours

- **Find** K-nearest neighbours of an input data
- **Count** class membership of the neighbours and find the majority class
- The majority class is the predicted class for the input



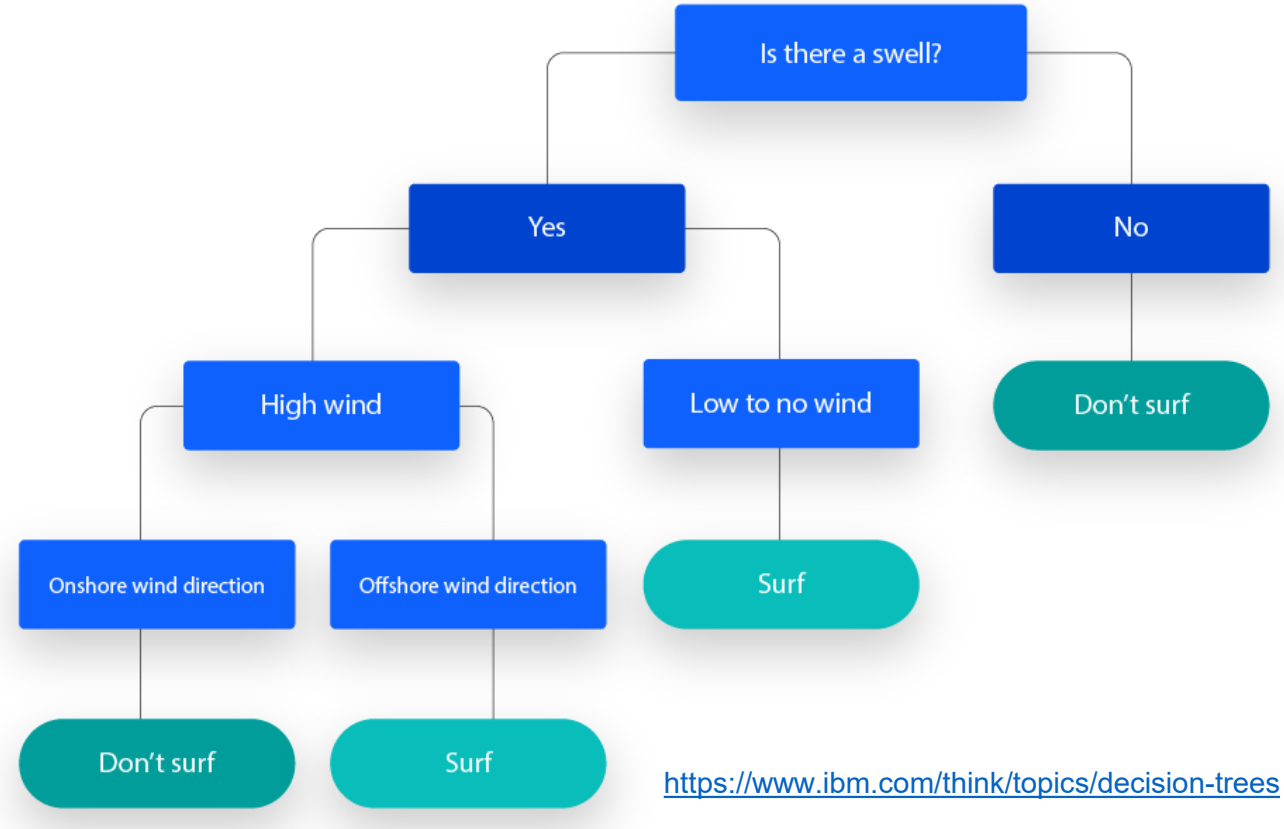
(a) 1-nearest neighbor (b) 2-nearest neighbor (c) 3-nearest neighbor

Predicted class for x according to 3-NN rule is +

For K-NN regression predict the average value of the neighbors

Decision Tree

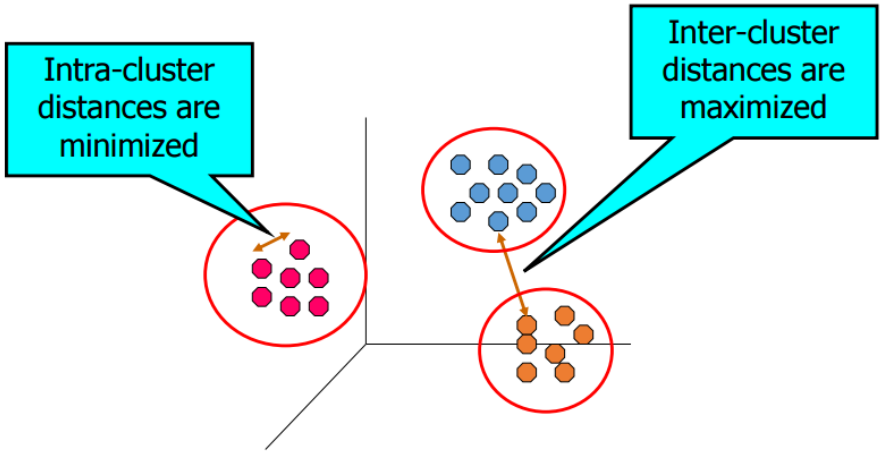
- **A decision tree** is a non-parametric supervised learning algorithm, which is utilized for both classification and regression tasks.
- It has a hierarchical, tree structure, which consists of a root node, branches, internal nodes and leaf nodes.



https://www.ibm.com/think/topics/decision-trees

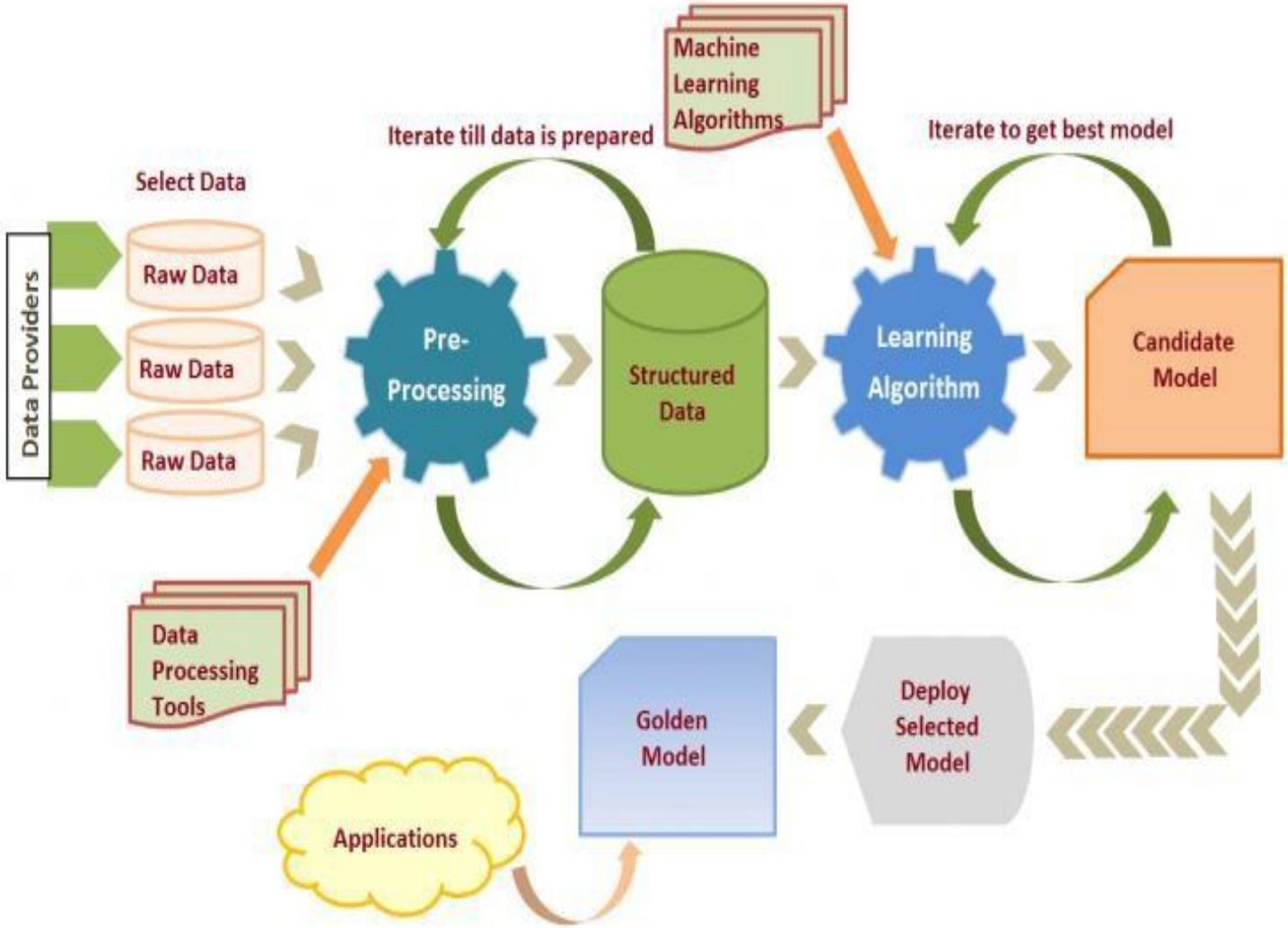
Clustering

- **Finding groups** of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



- **K-Means Clustering**
  - Partitional clustering approach
  - **Each cluster** is associated with a centroid (centre point)
  - **Each point** is assigned to the cluster with the closest centroid
  - Number of clusters, K, must be specified

AI: Machine Learning System Flow



AI: Transfer Learning

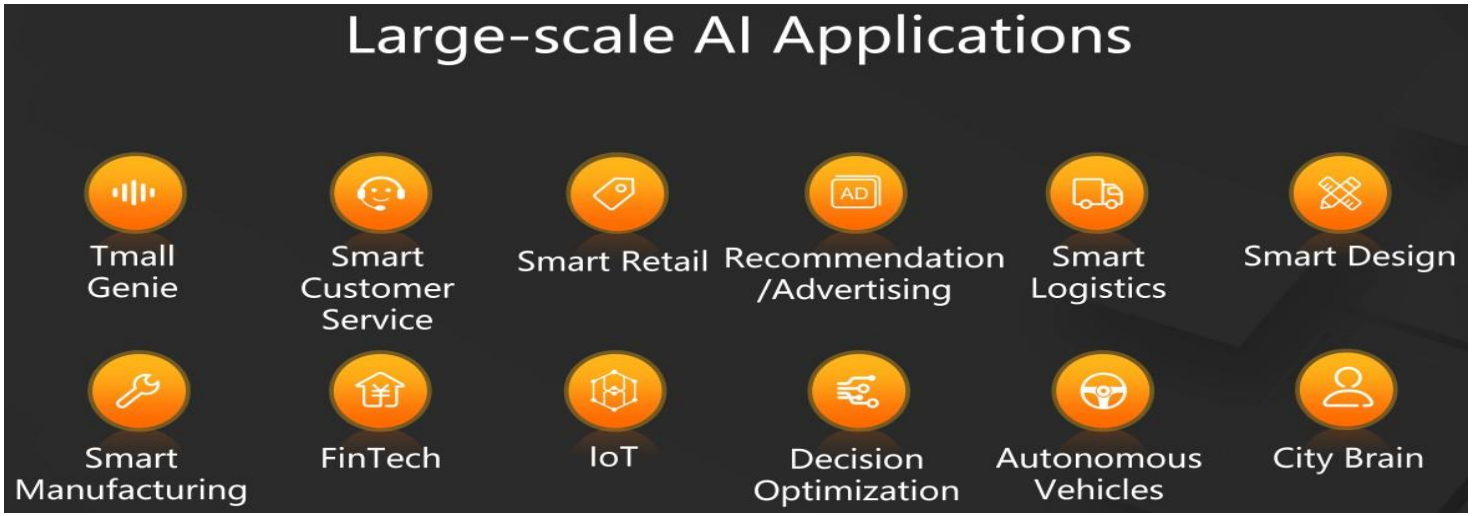
- **Definition**:
  - The ability of a system to use knowledge and skills learned from previous tasks to help with new tasks, especially when there are similarities between the tasks.
- **Motivation**:
  - Human learning inspires this concept—just like humans apply what they've learned to new situations.
- **Why Use Transfer Learning?**
  - Lack of Labelled Data: It's hard to find enough labelled data.
  - High Costs: Labelling data and calibrating models can be expensive.
  - Time-Consuming: The learning process can take a lot of time.

AI: Successful Applications

- Language Translation Services (Google)
- Translating Telephone (Skype)
- News Aggregation and Summarization (Google)
- Speech Recognition (Nuance)
- Song Recognition (Shazam)
- Face Recognition (Google)
- Image Classification/Recognition (Google)
- Question Answering (Apple Siri, IBM Watson, and etc.)
- Chess Playing (IBM Deep Blue)
- 3D Scene Modelling from Images (Microsoft Photosynth)
- Driverless Cars – Autonomous Driving (Google)
- Traffic Prediction (Inrix)

AI: Successful Applications

- **Cybersecurity**: Deep vulnerable code analysis, malware detection, intrusion detection, antispam, vulnerability management, normal and malicious data classification, insider attack prediction, adversarial AI (i.e. adversarial inputs, data poisoning, model stealing, and feedback weaponization), malware creation, smart botnets, spear phishing, conditional attacks, classify victims, security incident prediction, threat intelligence, and intelligent network networking.



AI in Engineering

- **Artificial Intelligence and Civil Engineering:**
  - Intelligent **detection** and **classification** of various types of defects in infrastructure surface images (cracks, deposit, etc.) can largely boost its maintenance efficiency.
  - Various supervised learning methods have been investigated for this task, including decision trees and support vector machines in previous studies, and deep neural networks more recently.

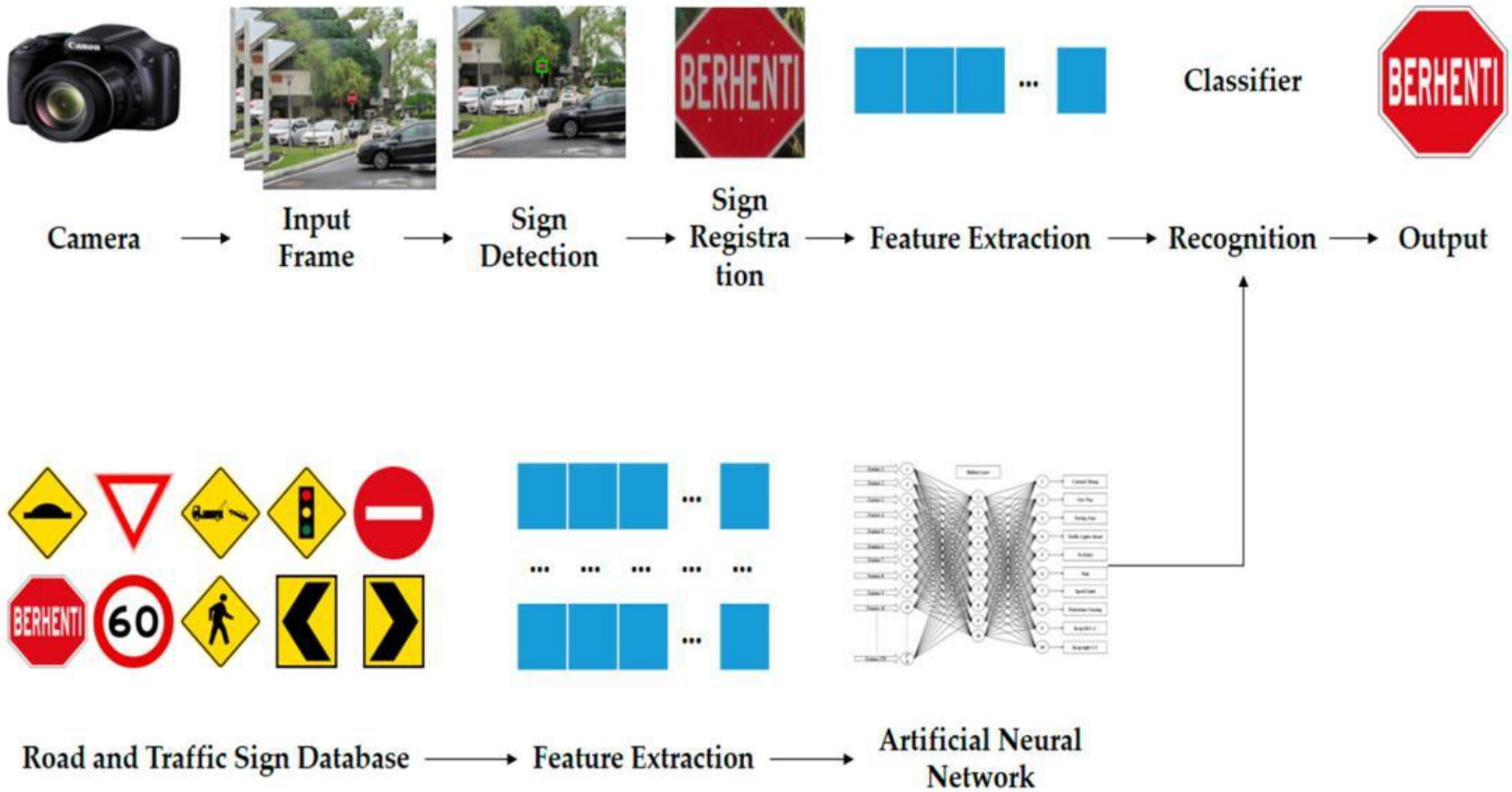
AI in Drug Discovery

- AI approaches provide a set of tools that can **improve** discovery and decision making for well-specified questions with abundant, and high-quality data.
- **The potential for various uses**, from initial screening of drug compounds to predicted success rate based on biological factors.
- Opportunities to apply AI occur in all stages of drug discovery.
- Examples include target validation, identification of prognostic biomarkers, and analysis of digital pathology data in clinical trials.

AI in Image Classification/Recognition

- **Image Processing to Computer Vision Steps**: Acquisition, Representation, Compression, Transmission, Image Enhancement, Edge/Feature Extraction, Pattern Matching, and Image Understanding or Recognition.
- **Definition**: The process of sorting pixels into a finite number of individual classes, or categories of data.
  - Based on their spectral response (i.e., the measured brightness of a pixel across the image bands, as reflected by the pixel's spectral signature).
- The underlying assumption of image classification is that spectral response of a particular feature (i.e., land-cover class) will be relatively consistent throughout the image.

AI Application in Image Classification/Recognition



AI Application in Cybersecurity (Hardware Security)

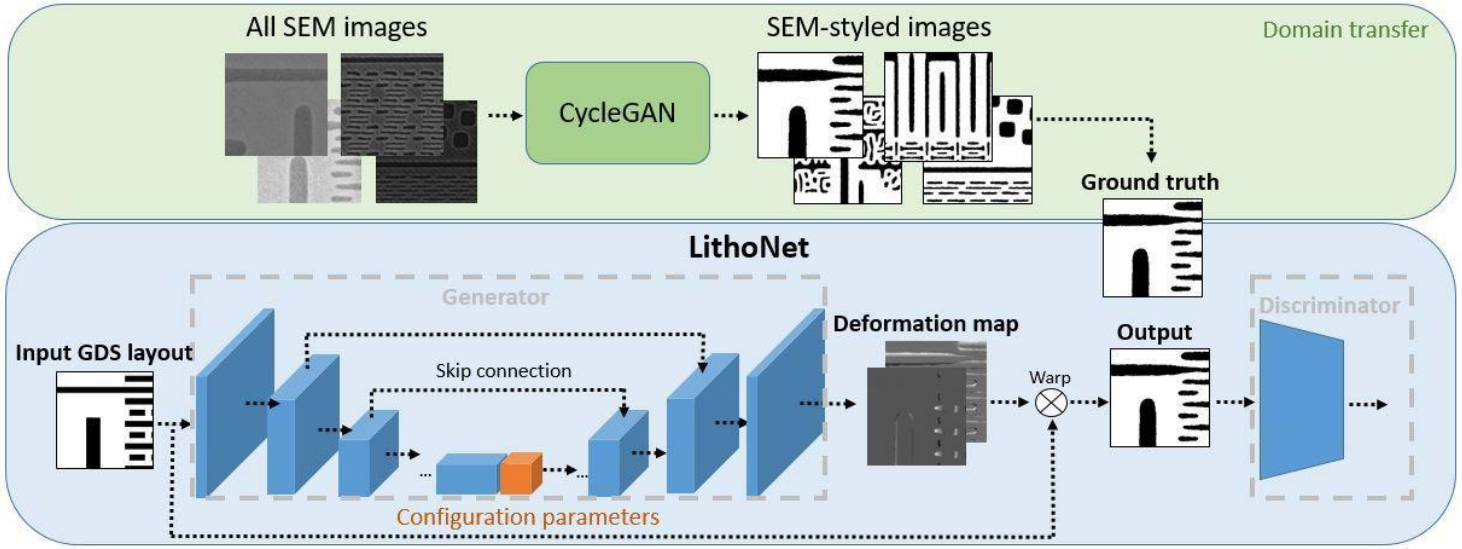
- AI techniques are applied to various hardware security problems.
- They can be incorporated both for attack and defense mechanisms.
- Possible defense candidate are Hardware Trojan Detection.
- Possible attack candidates are:
  - (a) side-channel analysis;
  - (b) launching modelling attacks on physically unclonable functions.

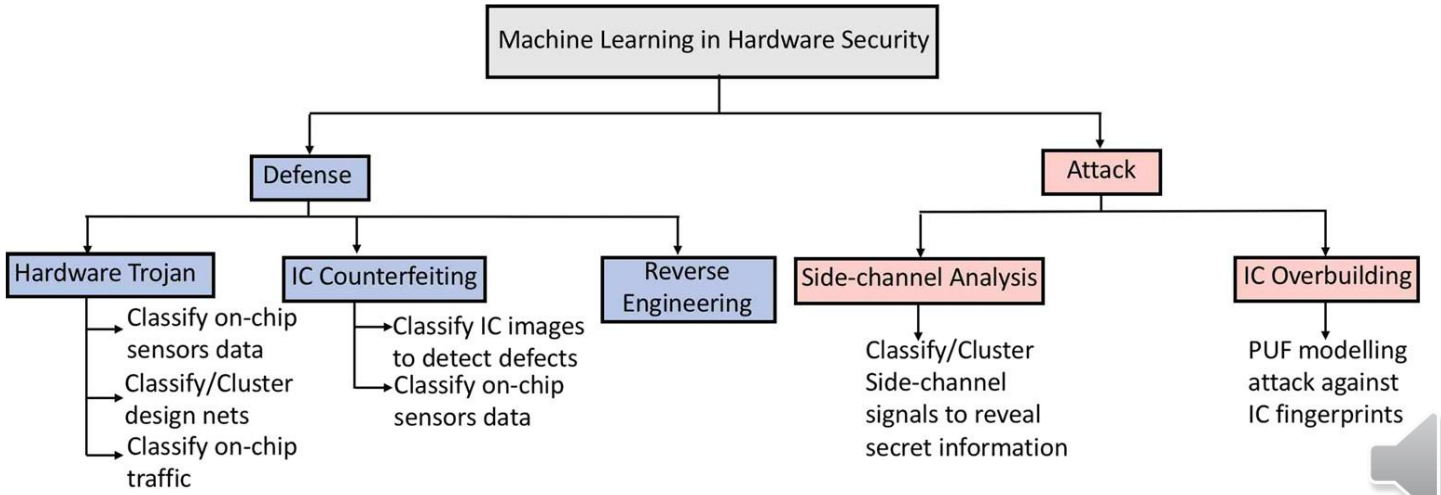
AI Application in Cybersecurity (Hardware Security)

- **Circuit Recognition**:
  - A deep learning framework using CNNs for recognizing circuit functionalities based on a new circuit representation suitable for network computing processes.

- **Physical Inspection of Electronics**:
  - Developing a CNN-based approach that learns the parametric model of physical and chemical phenomena of a fabrication process directly from a training dataset.
  - **This dataset** contains pairs of *Integrated Circuit* (IC) layouts and their corresponding *Scanning Electron Microscope* (SEM) images.
  - Based on the learned CNN model, we can predict a fabricated circuit shape and structure more accurately and efficiently than traditional methods along with evaluating the trustiness of the ICs.

AI Application in Cybersecurity (Hardware Security)





Summary

- In AI problems, we want to take an advantage of drawing a fairly sharp line between the physical and the intellectual capacities of a man.
- A knowledge-based agent steps:
  1) The stimulus must be translated into an internal representation;
  2) The representation is manipulated by cognitive processes to derive new internal representations;
  3) These is turn translated into action.
- Artificial neural network is computerized model of brain neural network.
- Processes in AI Systems: (a) representation; (b) thinking; (c) reasoning; (d) learning; (e) searching; and (f) output.
- AI systems can think and act like humans and also do them rationally.
- The intelligence as the computational part of the ability to achieve goals in the world.
- The intelligence is described as the capacity to learn and solve problems in particular tacking novel problems, act rationally, and act like humans.